CHAPTER TWO
LITERATURE REVIEW

2.0 Overview

This chapter presents a comprehensive review of literature related to the design and implementation of community incident reporting systems, with particular emphasis on geospatial technologies, web-based platforms, and citizen participation in emergency management. The purpose of this chapter is to examine existing studies, systems, and technological approaches relevant to the research problem identified in Chapter One.

The chapter is structured to first review related work in the domain of incident reporting systems and geospatial applications. It then analyzes existing tools and technologies used in such systems, highlighting their strengths and limitations. Furthermore, it establishes the relationship between previous studies and the proposed system. Finally, the chapter discusses the technologies and tools suitable for implementing the system and how these technologies can be effectively integrated.

Through this review, gaps in existing systems are identified, providing justification for the development of a more integrated, user-friendly, and geospatially enabled community incident reporting system.

2.1 Review of Related Work

Incident reporting plays a critical role in urban governance, public safety, and emergency management. Traditional reporting mechanisms such as phone calls, physical visits to offices, and informal communication channels are often inefficient, slow, and lack proper documentation. These approaches also suffer from poor location accuracy and limited accessibility, making it difficult for authorities to respond effectively.

With the advancement of digital technologies, several studies have proposed systems that address these limitations. Augustus et al. (2018) developed Project SIGAW, a community-based incident

reporting system integrated with Google Maps. The system allows users to submit reports that are visualized on a digital map, improving situational awareness and response coordination. This study demonstrated that integrating mapping technologies enhances the visibility of incidents and supports better decision-making.

Similarly, Bulusan et al. (2019) designed a mobile-based incident reporting application that utilizes GPS functionality to capture accurate location data. Their findings showed that mobile applications significantly reduce errors associated with manual location descriptions and improve response time. Nuevas et al. (2024) further extended this concept by developing a real-time incident reporting system that enables immediate communication between citizens and emergency responders. Their system highlighted the importance of real-time data transmission in improving emergency response efficiency.

In the domain of web-based systems, Arispe (2020) developed a road traffic incident response system that incorporates spatial data analysis. The study revealed that geospatial visualization tools enable authorities to identify patterns, prioritize incidents, and allocate resources more effectively. Similarly, Fitriani et al. (2019) created a GIS-based system for monitoring fire incidents, demonstrating the usefulness of spatial data in tracking incident distribution and identifying high-risk areas.

Anuar et al. (2025) proposed a web-based GIS system for traffic incident management, integrating mapping tools with route optimization features. Their system improved coordination among emergency responders and reduced response time. These studies collectively show that web-GIS platforms are essential for managing incident data and enhancing situational awareness.

Beyond individual systems, broader frameworks for integrating geospatial technologies into emergency management have been proposed. Bhanumurthy et al. (2015) developed a framework emphasizing the role of geospatial data in disaster preparedness, response, and recovery. Kawasaki et al. (2013) highlighted the increasing importance of web-based geospatial technologies in disaster management, particularly in enabling real-time information sharing and collaboration.

Community participation has also been recognized as a key factor in effective incident reporting. Apostol et al. (2023) explored a distributed system for real-time disaster reporting that incorporates community input. Their study emphasized the importance of citizen-generated data in improving situational awareness and resilience. Similarly, Zengeya et al. (2023) demonstrated that community-based reporting systems enhance transparency and accountability while improving communication between citizens and authorities.

Overall, the literature indicates that combining mobile technologies, web platforms, and geospatial tools can significantly improve incident reporting systems. However, many existing solutions are either limited in scope, lack integration, or are not tailored to specific community contexts.

2.1.1 Analysis of Existing Tools and Technologies

A variety of tools and technologies have been employed in the development of incident reporting systems. These technologies form the backbone of modern digital reporting platforms and enable efficient data collection, processing, and visualization.

One of the most widely used technologies is mobile application development with GPS integration. Mobile devices equipped with GPS sensors allow users to capture precise location data when reporting incidents. Studies by Bulusan et al. (2019) and Nuevas et al. (2024) show that GPS-enabled reporting significantly improves the accuracy and reliability of incident data.

Another key technology is web-based Geographic Information Systems (GIS). Web-GIS platforms enable the visualization of incident data on interactive maps, allowing users and authorities to analyze spatial patterns. Tools such as Google Maps API and other mapping frameworks are commonly used in such systems (Augustus et al., 2018; Fitriani et al., 2019).

Spatial databases are also essential for storing and managing geotagged data. Databases such as PostgreSQL with PostGIS extension or MySQL with spatial capabilities allow efficient storage,

querying, and analysis of spatial information. These databases support features such as proximity analysis and clustering, which are useful for identifying incident hotspots.

In addition, cloud computing technologies play a significant role in modern systems. Cloud platforms provide scalability, real-time data synchronization, and accessibility across multiple devices. This ensures that users and authorities can access the system anytime and from anywhere.

Another important component is notification and communication systems, such as publish/subscribe mechanisms. Kassab et al. (2010) demonstrated that such systems enable real-time alerts and updates, improving situational awareness during emergencies.

Despite the availability of these technologies, their implementation varies across systems. Some systems focus primarily on mobile reporting, while others emphasize web-based visualization. Few systems achieve seamless integration of all these components into a unified platform.

2.1.2 Analysis of Strengths and Weaknesses

Strengths of Existing Systems

Existing incident reporting systems offer several advantages. One major strength is improved location accuracy through GPS-enabled reporting, which reduces ambiguity and enhances response efficiency (Bulusan et al., 2019).

Another strength is real-time communication, which allows incidents to be reported and addressed promptly (Nuevas et al., 2024). This is particularly important in emergency situations where delays can have serious consequences.

Geospatial visualization is another significant benefit. GIS tools enable authorities to identify patterns, trends, and hotspots, supporting data-driven decision-making (Fitriani et al., 2019).

Furthermore, community-based systems promote citizen engagement and transparency. By allowing users to report incidents and track their status, these systems foster trust and collaboration between citizens and authorities (Augustus et al., 2018).

Weaknesses of Existing Systems

Despite their advantages, existing systems have several limitations. One common issue is the lack of integration. Many systems focus on specific types of incidents, such as traffic or fire, rather than providing a comprehensive solution for all community incidents.

Usability is another challenge. Some systems are complex and difficult for non-technical users to navigate, limiting their adoption.

Infrastructure dependency is also a concern. The performance of these systems often depends on internet connectivity and device capabilities, which may vary across users (Fitriani et al., 2019).

Data reliability is another limitation, as user-generated reports may contain inaccurate or misleading information (Apostol et al., 2023).

Finally, many systems are developed as prototypes and lack scalability, security, and long-term sustainability features required for real-world deployment.

2.1.3 Relation to the Proposed System

The reviewed literature provides a strong foundation for the proposed system. While existing systems demonstrate the effectiveness of geospatial technologies and digital reporting platforms, they also reveal significant gaps that need to be addressed.

The proposed system aims to bridge these gaps by developing a unified platform that integrates mobile and web technologies with geospatial capabilities. Unlike many existing systems that focus on specific incident types, the proposed system will support a wide range of community incidents, making it more versatile and applicable to real-world scenarios.

Additionally, the system will emphasize usability by providing an intuitive interface that can be easily used by individuals with varying levels of technical expertise. It will also incorporate features such as status tracking and feedback mechanisms to enhance transparency and user engagement.

Furthermore, the proposed system will include basic spatial analysis features such as clustering and hotspot detection, enabling authorities to make informed decisions. By addressing the limitations identified in existing systems, the proposed solution aims to provide a more comprehensive, efficient, and user-centered approach to community incident reporting.

2.2 Technologies and Tools

The development of the proposed system requires the integration of multiple technologies. These technologies are selected based on their ability to support system functionality, scalability, and usability.

Frontend technologies such as React.js which is a JavaScript framework will be used to develop the user interface. These technologies enable the creation of responsive and interactive interfaces that enhance user experience.

Backend technologies such as Laravel which is a PHP framework will handle server-side processing, including data validation, authentication, and communication with the database.

Database systems such as MySQL will be used to store incident data. When combined with spatial extensions, these databases can efficiently manage geospatial data.

Geospatial tools such as Google Maps API will be used for map integration and visualization. These tools provide features such as markers, geocoding, and route mapping.

Cloud services will be used for hosting and scalability, ensuring that the system can handle multiple users and large volumes of data.

2.2.1 Integration of Technologies

The success of the proposed system depends on the effective integration of its components. The system will adopt a layered architecture consisting of presentation, application, data, and geospatial layers.

The presentation layer will provide the user interface for both citizens and authorities. The application layer will handle business logic and communication between components. The data layer will store incident data, while the geospatial layer will handle mapping and spatial analysis.

For example, when a user reports an incident, the system captures the GPS coordinates and sends the data to the server. The server processes and stores the data in the database, and the geospatial layer displays the incident on a map. Authorities can then analyze the data and respond accordingly.

This integration ensures seamless data flow, real-time updates, and efficient system performance.

2.3 Chapter Summary

This chapter reviewed existing literature on community incident reporting systems, geospatial technologies, and related tools. It examined previous studies, analyzed existing technologies, and identified strengths and weaknesses of current systems. The chapter also established the relevance of these studies to the proposed system and discussed the technologies required for implementation.

The review highlights the need for a comprehensive, integrated, and user-friendly system that leverages geospatial technology to improve incident reporting and management. The next chapter will present the system design and methodology used to develop the proposed solution.

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